

Emotion Guided Speech-Driven Facial Animation

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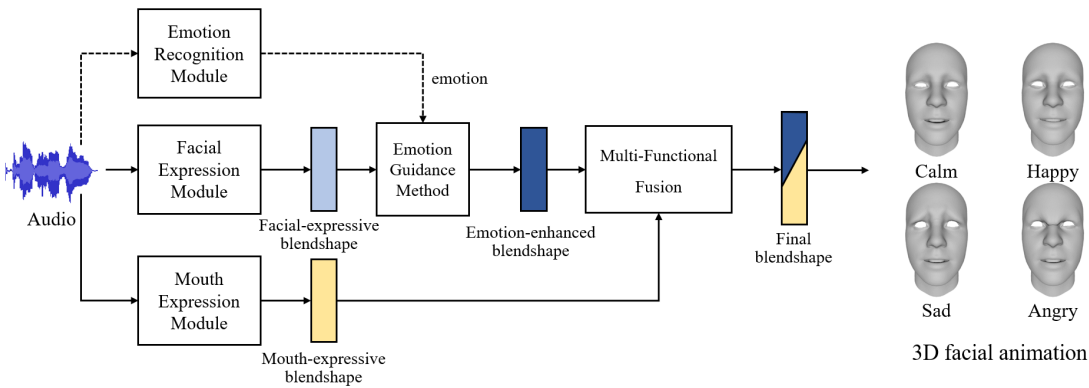


Figure 1: An illustration of Emotion Guided Speech-Driven Facial Animation pipeline

ABSTRACT

The modern deep neural network has allowed an applicable level of speech-driven facial animation, simulating natural and precise 3D animation from speech data. Regardless, many of the works show weakness in drastic emotional expression and flexibility of the animation. In this work, we introduce emotion guided speech-driven facial animation, simultaneously proceeding with classification and regression from the speech data to generate a controllable level of evident emotional expression on facial animation. Performance using our method shows reasonable expressiveness of facial emotion with controllable flexibility. Extensive experiments indicate that our method generates more expressive facial animation with controllable flexibility compared to previous approaches.

KEYWORDS

speech-driven facial animation, emotion recognition, audio processing, deep neural networks

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1 INTRODUCTION

Conventionally in the 3D game industry, designers manually hand-craft the animation for each frame, costing an extensive amount of labor and time. Development in deep neural networks (DNN) has allowed speech-driven facial animation (SDFA) to generate favorable quality of automation in such a process, focusing on producing precise and realistic facial expressions in 3D animation. However, the reflection of emotional elements to fabricate a drastic facial expression has not yet been deeply studied, while the emotional elements contain potentials for a more favorable quality of the facial expression.

A Study done by Karras et al. [Karras et al. 2017] was one of the first to produce an outstanding SDFA result, simulating realistic facial animation based on DNN with data from real human expression. MulTaNet [Kim et al. 2019] has developed SDFA on multiple characters by multiple feature learning and character-dependent prediction, improving the flexibility of the network. While these methods suffered from preciseness in mouth movement, VOCA

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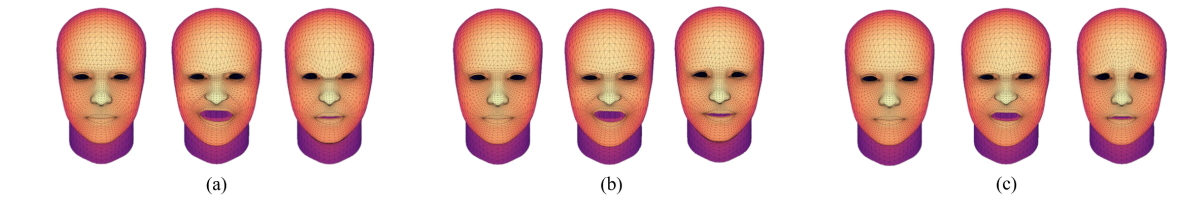


Figure 2: Comparison with previous methods on various emotional expressions: angry (a), happy (b), and sad (c). Each expression includes results from VOCA [Cudeiro et al. 2019] (left), the LSTM [Pham et al. 2018] (middle), and ours (right).

[Cudeiro et al. 2019] was introduced to improve the quality of mouth animation with an accurate movement.

While many studies aim to simulate realistic high-quality preciseness in facial expression, they meet the limitations in the robustness and flexibility in various emotional expressions, which becomes a highly desired application in the 3D game industry.

To this end, we introduce an emotion guided SDFA framework named EG-SDFA, which leverages the results from multiple DNNs predictions to generate the emotion-enhanced facial animation and yet very accurate lip synchronization.

2 EG-SDFA

Our approach solves the emotional expression as a combination of classification and regression, consist of an emotion recognition network (ERN), a facial expression network (FEN), and a mouth expression network (MEN). With emotion-guided projection, ERM and FEM create emotion-enhanced blendshapes, which are integrated with the mouth-expressive blendshapes from MEN via multi-functional fusion.

For the emotion-guided projection, the FEN first predicts the blendshapes for the facial expression of the given speech, followed by emotion classification by ERN. With the classified emotion, a corresponding master blendshape is loaded, that its unit vector is used for the linear interpolation with the unit vectors of the predicted blendshapes. Finally, the predicted blendshapes are projected onto the interpolated vector, forming emotion-enhanced blendshapes.

Assuming that the master blendshape represents an ideal condition of its emotional expression, such a method allows each prediction to reflect emotional features by following the master's guidance. Each master blendshape can be predefined as an average of the target data from the training of FEN or custom generated.

To simulate precise mouth animation in the emotion-enhanced blendshapes, we structure multi-functional fusion to leverage from a network of specialty in mouth expression. For a raw combination of the mouth-expressive blendshape with the emotion-enhanced blendshape produces awkward features, multiple parametric functions to mitigate the deficiencies and to enhance the quality are included. To combine the two types of blendshape, we perform linear weighted summation by assigning an importance weight for each. The intensity control of the emotion is done by scaling the blendshapes. Linear multiplication has shown to be an effective way to control the intensity monotonously, while non-linear gamma transformation has been shown to change the ratio of the blendshapes to simulate non-uniform variation in emotional expression.

In order to control the dynamic range, we calculate the magnitude of blendshapes for each frame and manipulates the distribution, with Gaussian assumption. The modified magnitude values are re-assigned to the blendshape, resulting in flexible control in the range of movement. To control the smoothness of the animation, we implement 1-D Gaussian blurring and high boost filtering iteratively to simulate smooth or "shaky" transition. For more implementation details, please refer to our supplementary materials.

3 EXPERIMENT

In EG-SDFA, we use a multi-layer perceptron for ERN, an LSTM network [Pham et al. 2018] for FEN, VOCA [Cudeiro et al. 2019] for MEN. For the demonstration, speech data from RAVDESS dataset [Swanson et al. 2019] was used to produce 8 different emotional expressions. Performance from various emotional speech indicates that our method generates evident emotional expressions (Figure 2) yet accurate mouth animation. Extensive ablation studies of multi-functional fusion show the flexible capability to control the result.

4 CONCLUSION

In this work, we proposed EG-SDFA to simulate controllable and emotionally expressive facial animation from given speech data. By leveraging and overcoming the limitations of the previous methods, our EG-SDFA generates more realistic facial animation with expressive emotions and accurate mouth movement. EG-SDFA is already integrated into 3D mobile game resource production pipeline.

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